No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

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Abstract

In a *repeated Stackelberg Security Game (SSG)*, a defender commits each round to a randomized deployment of its resources and an adversarial sequence of attackers, each drawn from some set K of *types*, best-responds. Balcan et al. [2015] designed no-regret online learning algorithms for this model under the assumption that the set K is *known to the defender in advance*, achieving expected regret $O(\sqrt{Tn^2k\log(nk)})$ against an oblivious adversarial sequence. They also show a matching lower bound: linear-in-T regret is unavoidable when K is unrestricted. In practice, however, the defender rarely knows K. We relax the known-K assumption to the natural middle ground $|K| \le K_{\max}$, where the bound $K_{\max}$ is known but the actual types are discovered online. We propose Grow-Catalogue, an algorithm that maintains the polytope partition of the coverage space induced by the attacker types seen so far and re-runs an online learning sub-routine whenever a new type is observed. We prove that under full- information feedback Grow-Catalogue achieves expected regret

$$O\Big(\sqrt{T \cdot K_{\max} \cdot \big(n^2 + K_{\max}\log n\big)}\Big),$$

matching the bound of Balcan et al. [2015] (with $k \mapsto K_{\max}$) up to constants. The bound is meaningful as long as $K_{\max} \cdot (n^2 + K_{\max}\log n) \ll T$; beyond this regime it is matched by the linear lower bound from the unbounded side. We validate the theorem with experiments on synthetic SSG instances under a staggered-uniform adversary, observing the predicted sublinear behaviour and clear spikes in per-round regret at each discovery round. Our implementation is available at github.com/CS6002-SSG/grow-catalogue.

1 Introduction

A *Stackelberg Security Game* (SSG) models the strategic interaction between a *defender* protecting n targets with limited resources and an *attacker* who chooses a target to attack. The defender commits to a mixed strategy; the attacker observes the commitment and best-responds. Computational tools built around this model are deployed at, among others, the US Coast Guard, LAX airport, the Federal Air Marshals service, and wildlife anti-poaching programs [Tambe, 2012, Yang et al., 2014].

All such deployments require knowledge of the attacker's utility function. In practice the defender faces many possible attacker types whose utilities are only partially known. A natural online-learning formulation, due to Balcan et al. [2015], treats the sequence of attackers as *adversarial*: at round t an attacker of some type $a_t \in K$ arrives, and the defender must maintain low regret against the best fixed commitment in hindsight. Balcan et al. [2015] show that for $|K| = k$ known and full-information feedback, a polynomial-weights algorithm over the ϵ-approximate extreme points of the best-response polytope partition achieves expected regret $O(\sqrt{Tn^2k\log(nk)})$. They also prove that no online algorithm can avoid $\Omega(T)$ regret in the fully unrestricted setting.

Motivation. The assumption that K is known in advance is strong. A realistic scenario is that the defender has domain knowledge of an *upper bound* $K_{\max}$ on the number of distinct attacker types it might face (e.g., "at most five distinct threat actors"), but the utilities of any particular attacker type are only revealed the first time that type attacks. In this paper we ask:

Can sublinear regret be achieved when the set of attacker types K is unknown a priori but satisfies $|K| \le K_{\max}$?

This question sits cleanly in the gap between the "known K" setting and the negative result for unrestricted K in Balcan et al. [2015].

Contributions. Our contributions are as follows.

1. *Problem formulation.* We formalize the unknown-K variant of the repeated Stackelberg security game (Section 4).
2. *Algorithm.* We design Grow-Catalogue (Algorithm 1), which at each round maintains a catalogue C_t of discovered attacker types, recomputes the set $\mathcal{E}(C_t;\epsilon)$ of ϵ-approximate extreme points of the partition of $\mathcal{P}$ induced by C_t, and runs an anytime multiplicative- weights sub-routine over $\mathcal{E}(C_t)$. When a new type is observed, the catalogue is updated and the sub-routine is restarted on the enlarged expert set.
3. *Regret bound.* We prove (Theorem 7) that Grow-Catalogue has expected regret $O(\sqrt{TK_{\max}(n^2 + K_{\max}\log n)})$. This matches Balcan et al.'s Theorem 5.1 with k replaced by $K_{\max}$; for constant $K_{\max}$ the rate is $\tilde{O}(n\sqrt{T})$.
4. *Regime of applicability.* We characterize the range of $K_{\max}$ for which the bound is non-trivial (Corollary 8) and show that the instance-adaptive refinement depends on the number K' of types *actually* observed, not on $K_{\max}$.
5. *Experiments.* We implement Grow-Catalogue with two variants of the polytope recomputation (total and incremental) and evaluate them on synthetic SSG instances under a *staggered-uniform* adversary. The measured regret is sublinear, with visible spikes at rounds of first appearance of each new type, confirming the epoch structure exploited by the proof (Section 6).

Organization. Section 2 surveys related work. Section 3 recalls the Stackelberg security game formalism of Balcan et al. [2015]. Section 4 states the unknown-K problem. Section 5 presents the algorithm, states the main regret theorem, and gives the complete proof. Section 6 reports the experiments. Section 7 concludes.

2 Related Work

The line of work most closely related to ours is online learning for SSGs with *attacker uncertainty*.

Prior-based approaches. Pita et al. [2010] address uncertainty by solving a robust optimization problem against worst-case attacker payoffs. Kiekintveld et al. [2011] and Marecki et al. [2012] study *Bayesian* SSGs in which a prior distribution over attacker types is assumed; their algorithms require this prior, which is usually estimated from historical data and hard to validate.

Learning a single unknown type. Letchford et al. [2009] give a learning algorithm that, through best-response queries, identifies the attacker's utility function with polynomial query complexity. Blum et al. [2014] extend this to an "adaptive commitment" setting in which the defender refines its strategy round-by-round against a single fixed, but unknown, type.

Adversarial sequence of types. Balcan et al. [2015] depart from the single-type assumption. They model the sequence of attackers as adversarially chosen from a *known* finite set K and design no-regret algorithms for both full-information and partial- information (bandit) feedback. Their full-information bound, $O(\sqrt{Tn^2k\log(nk)})$, is the starting point of our work. They also prove an $\Omega(T)$ lower bound when K is allowed to grow with T.

Online learning with changing expert sets. The internal sub-routine of Grow-Catalogue is an anytime variant of the Polynomial Weights / Hedge algorithm [Littlestone and Warmuth, 1994, Freund and Schapire, 1997, Cesa-Bianchi and Lugosi, 2006], which we restart each time the catalogue changes. More sophisticated mechanisms for growing expert sets exist [Mourtada and Maillard, 2017, Gaillard et al., 2019]; they shave logarithmic factors but do not change the leading-order rate, and we stick with the simpler restarting strategy.

Other related directions. Haghtalab et al. [2022] study no-regret learning against non-myopic (strategic) agents in a single-type query model. Harris et al. [2024] add *side information*: contexts observed by both players. Neither addresses the "K unknown but bounded" question we pose here. To the best of our knowledge no prior work combines the adversarial-types setting of Balcan et al. [2015] with an unknown K of bounded size, although the gap between Balcan et al.'s upper and lower bounds clearly invites this extension.

3 Preliminaries

We recall the formal model of Balcan et al. [2015, §2], adopting their notation throughout.

Game primitives. A repeated SSG has a time horizon $T \in \mathbb{N}$, a set of targets $N = \{1, \dots, n\}$, a defender with R resources, and a collection $\mathcal{D} \subseteq 2^N$ of *schedules*, each of which represents a set of targets a single resource can defend simultaneously. An *assignment function* $A : R \to 2^{\mathcal{D}}$ indicates the set of schedules each resource may be assigned to. A *pure strategy* is any valid resource-to-schedule assignment. Let m denote the number of pure strategies, and let M be the $\{0,1\}$-valued $n \times m$ indicator matrix with $M_{i,j} = 1$ iff target i is covered under the j-th pure strategy. A *mixed strategy* is a distribution $\mathbf{s} \in \Delta_m$ and its induced *coverage probability vector* is $\mathbf{p} = M\mathbf{s} \in [0,1]^n$, where p_i is the probability that target i is defended. The set $\mathcal{P} \subseteq [0,1]^n$ of valid coverage probability vectors is a convex polytope.

Payoffs. Each target i carries defender utilities $u_d^c(i)$ (if i is attacked and covered) and $u_d^u(i)$ (attacked and uncovered), both in $[-1,1]$. The defender's *expected* utility when target i is attacked under coverage $\mathbf{p}$ is

$$U_d(i, \mathbf{p}) \;=\; u_d^c(i)\, p_i + u_d^u(i)\,(1 - p_i),$$

which is linear in $\mathbf{p}$ for fixed i.

Attacker types. An attacker *type* α is a pair of utility vectors $u_\alpha^c, u_\alpha^u \in [-1,1]^n$; its expected utility from attacking target i under coverage $\mathbf{p}$ is $U_\alpha(i,\mathbf{p}) = u_\alpha^c(i)p_i + u_\alpha^u(i)(1-p_i)$. Observing $\mathbf{p}$, type α *best-responds* by attacking

$$b_\alpha(\mathbf{p}) \;=\; \arg\max_{i \in N} U_\alpha(i, \mathbf{p}), \tag{1}$$

with ties broken arbitrarily but consistently.

Repeated game and regret. There is a (possibly unknown) set K of attacker types. An *oblivious* adversary fixes a sequence $\mathbf{a} = a_1, \dots, a_T \in K^T$ before the game starts. At each round t the defender commits $\mathbf{p}_t \in \mathcal{P}$, attacker a_t best-responds, and the defender receives $U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)$. The defender's benchmark is the best fixed coverage vector in hindsight,

$$\mathbf{p}^* \;=\; \arg\max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T} U_d\big(b_{a_t}(\mathbf{p}), \mathbf{p}\big), \tag{2}$$

and the (expected) regret of an online algorithm is

$$\mathrm{Regret}(T) \;=\; \sum_{t=1}^{T} U_d\big(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*\big) - \mathbb{E}\Big[\sum_{t=1}^{T} U_d\big(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t\big)\Big]. \tag{3}$$

Polytope structure. The following two lemmas of Balcan et al. [2015] are the technical machinery we will inherit. For a set of types $C \subseteq K$, a type $\alpha \in C$ and a target $i \in N$, define

$$\mathcal{P}_i^{\alpha}(C) \;=\; \big\{\mathbf{p} \in \mathcal{P} \,:\, b_\alpha(\mathbf{p}) = i\big\}, \tag{4}$$

a convex polytope defined by the pairwise linear inequalities $U_\alpha(i, \mathbf{p}) \geq U_\alpha(j, \mathbf{p})$ for all $j \neq i$. For a response profile $\sigma : C \to N$ let $\mathcal{P}_\sigma(C) = \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^{\alpha}(C)$; the non-empty $\mathcal{P}_\sigma(C)$ partition $\mathcal{P}$ into convex regions in which every $\alpha \in C$ has a fixed best response. Within each region the defender's payoff as a function of $\mathbf{p}$ is *linear*.

Lemma 1 (Balcan et al., 2015, Lem. 4.3)**.** *Fix $\epsilon > 0$. There exists a finite set $\mathcal{E}(C;\epsilon) \subseteq \mathcal{P}$ (the ϵ-approximate extreme points of the partition induced by C) such that for any sequence $\mathbf{a}'$ of attackers from C of length T',*

$$\max_{\mathbf{p} \in \mathcal{E}(C;\epsilon)} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) \geq \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - 2\epsilon T'.$$

Lemma 2 (Balcan et al., 2015, Lem. 4.4)**.** *The set $\mathcal{E}(C;\epsilon)$ of Lemma 1 can be chosen so that $|\mathcal{E}(C;\epsilon)| \in O\big((2^n + |C|n^2)^n \cdot n^{|C|}\big)$.*

Proposition 3 (Balcan et al., 2015, Prop. 1)**.** *For any fixed expert set $\mathcal{M}$ and any loss sequence in $[-\kappa, \kappa]$, there is an online learning algorithm whose expected regret after T' rounds is at most $O\big(\sqrt{T'\kappa \log|\mathcal{M}|}\big)$, without requiring T' to be known in advance.*

Proposition 3 is achieved, e.g., by Polynomial Weights / Hedge with a time-varying learning rate $\eta_t = \sqrt{\log|\mathcal{M}|/t}$, which gives an *anytime* guarantee; see Cesa-Bianchi and Lugosi [2006, §2].

4 Unknown K with a Known Bound $K_{\max}$

We depart from Balcan et al. [2015] in exactly one point: the set K is not revealed to the defender at the start.

Problem 4 (Unknown-K repeated SSG)**.** A repeated SSG as in Section 3 is played, with the following two changes.

1. The set K of attacker types is *unknown* to the defender at the start, but $|K| \leq K_{\max}$ for a known constant $K_{\max}$.
2. Feedback is *full information*: at the end of round t the defender observes a_t, and the first time any type α is observed, its utility vectors (u_α^c, u_α^u) are revealed.

The defender's benchmark and regret are still those of Equations (2) and (3).

Remark 5 (Necessity of a bound on $|K|$). Balcan et al. [2015, Thm. 7.1] show that if $|K| < 2^{T+1}$ is unrestricted, every online algorithm has regret $\Omega(T)$. Hence *some* bound on $|K|$ is needed for a sublinear guarantee; our assumption of a known constant $K_{\max}$ is the least restrictive such bound.

Remark 6 (Benchmark is well-defined). The benchmark $\mathbf{p}^*$ of Equation (2) depends only on the realized sequence $\mathbf{a}$, which contains at most $K_{\max}$ distinct types. So even though K is unknown to the defender, $\mathbf{p}^*$ itself is always a single fixed element of $\mathcal{P}$.

5 Algorithm and Analysis

5.1 The Grow-Catalogue Algorithm

At any round t, the defender holds a *catalogue* $C_t \subseteq K$ of types observed so far and the associated expert set $\mathcal{E}_t = \mathcal{E}(C_t;\epsilon)$, where $\epsilon = 1/\sqrt{T}$. An anytime Polynomial-Weights instance is run over $\mathcal{E}_t$. When a new attacker type is observed, C_t is enlarged, $\mathcal{E}_t$ is recomputed, and the sub-routine is restarted on the new expert set. Algorithm 1 gives the pseudocode.

Algorithm 1 GROW-CATALOGUE (full information)

Require: Horizon T; bound $K_{\max}$; game primitives; default coverage $\mathbf{p}_{\text{def}}$ (e.g. uniform)

1: $C \leftarrow \emptyset$, $\mathcal{E} \leftarrow \emptyset$, $\epsilon \leftarrow 1/\sqrt{T}$, $t_{\text{loc}} \leftarrow 0$
2: **for** $t = 1, \ldots, T$ **do**
3: **if** $\mathcal{E} = \emptyset$ **then**
4: play $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$
5: **else**
6: sample $\mathbf{p}_t \sim q_t$ from the current PW distribution over $\mathcal{E}$
7: **end if**
8: observe a_t
9: **if** $a_t \notin C$ **then** ▷ new type discovered
10: $C \leftarrow C \cup \{a_t\}$; record $(u_{a_t}^c, u_{a_t}^u)$
11: $\mathcal{E} \leftarrow \mathcal{E}(C;\epsilon)$ ▷ recompute partition
12: restart PW on $\mathcal{E}$; $t_{\text{loc}} \leftarrow 0$
13: **else** ▷ routine update
14: $t_{\text{loc}} \leftarrow t_{\text{loc}} + 1$
15: $\ell_t(\mathbf{p}) \leftarrow -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$ for each $\mathbf{p} \in \mathcal{E}$
16: $\eta \leftarrow \sqrt{\log|\mathcal{E}|/t_{\text{loc}}}$
17: $w_{\mathbf{p}} \leftarrow w_{\mathbf{p}}\, e^{-\eta\, \ell_t(\mathbf{p})}$ for each $\mathbf{p} \in \mathcal{E}$
18: **end if**
19: **end for**

We implement two variants of line 11: (i) *total recomputation*, which rebuilds $\mathcal{E}(C;\epsilon)$ from scratch, and (ii) *incremental*, which reuses PW weights for extreme points that survive the new refinement of the partition. The two variants are observationally equivalent – they achieve *identical* regret – but the incremental variant has lower per-discovery wall-clock cost.

5.2 Main Result

Theorem 7 (Unknown K, full information)**.** *Let K be an unknown set of attacker types with $|K| \leq K_{\max}$, and fix a horizon T. Under full-information feedback and for any oblivious adversarial sequence $\mathbf{a} \in K^T$, Algorithm 1 satisfies*

$$\mathbb{E}[\text{Regret}(T)] \;\leq\; O\Big(\sqrt{T\, K_{\max}\,(n^2 + K_{\max}\log n)}\Big). \tag{5}$$

In particular, if $K_{\max} = O(1)$ then $\mathbb{E}[\text{Regret}(T)] = \tilde{O}(n\sqrt{T})$, matching Balcan et al. [2015, Thm. 5.1] up to logarithmic factors.

The proof of Theorem 7 is the central technical contribution of this paper, and we give it in full.

Proof of Theorem 7. Fix the (unknown) adversarial sequence $\mathbf{a} = a_1, \ldots, a_T \in K^T$ and let $K' \leq K_{\max}$ denote the number of *distinct* attacker types that appear in $\mathbf{a}$. We organize the proof in seven steps.

Step 1: Epoch decomposition. Let $\tau_1 < \tau_2 < \cdots < \tau_{K'}$ be the round indices at which each new type is first observed (so a_{τ_i} is seen for the first time at round τ_i, and $\tau_1 \geq 1$). Define, for $i = 1, \ldots, K'$,

$$\mathrm{Ep}_i \;=\; \{\tau_i + 1, \tau_i + 2, \ldots, \tau_{i+1} - 1\},$$

with the convention $\tau_{K'+1} = T + 1$. Let $L_i = |\mathrm{Ep}_i| = \tau_{i+1} - \tau_i - 1$. Let $\mathcal{D} = \{\tau_1, \ldots, \tau_{K'}\}$ be the set of *discovery rounds*. Then $[1, T] = \mathcal{D} \sqcup \bigcup_{i=1}^{K'} \mathrm{Ep}_i$, and $|\mathcal{D}| + \sum_{i=1}^{K'} L_i = T$, with $|\mathcal{D}| = K' \leq K_{\max}$.

Step 2: Catalogue is fixed inside an epoch. After round τ_i the catalogue is $C_i = \{a_{\tau_1}, \ldots, a_{\tau_i}\}$, and it remains unchanged throughout Ep_i. Moreover, every round $t \in \mathrm{Ep}_i$ has $a_t \in C_i$: by definition the next *new* type arrives only at round τ_{i+1}, so any a_t with $t < \tau_{i+1}$ is already in C_i. Hence during Ep_i the algorithm runs a single, fresh Polynomial-Weights instance on the expert set $\mathcal{E}_i := \mathcal{E}(C_i; \epsilon)$ against a sequence of attackers all of which live in C_i.

Step 3: Within-epoch PW regret. By Proposition 3 applied to the PW instance of Algorithm 1 running on $\mathcal{E}_i$ over the L_i rounds of Ep_i (the loss range is $\kappa = 1$ because $U_d \in [-1, 1]$), there is an absolute constant $c_1 > 0$ such that

$$\max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \mathrm{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E}\left[\sum_{t \in \mathrm{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \;\leq\; c_1 \sqrt{L_i \log |\mathcal{E}_i|}, \tag{6}$$

where $\mathbf{p}_t \sim q_t$ is the random strategy played at round t under the PW distribution q_t on $\mathcal{E}_i$, and the expectation is over the algorithm's internal randomness. The anytime learning rate $\eta_{t_{\mathrm{loc}}} = \sqrt{\log |\mathcal{E}_i| / t_{\mathrm{loc}}}$ used in Algorithm 1 is what makes the bound Equation (6) hold without requiring the epoch length L_i to be known in advance [Cesa-Bianchi and Lugosi, 2006, §2].

Step 4: Within-epoch approximation via Lemma 1. The sub-sequence $(a_t)_{t \in \mathrm{Ep}_i}$ has length L_i and uses only types in C_i, which is precisely the hypothesis of Lemma 1 applied to $C = C_i$. Hence

$$\max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \mathrm{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \geq \max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \mathrm{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - 2\epsilon L_i. \tag{7}$$

Since $\mathbf{p}^* \in \mathcal{P}$, the right-hand side of Equation (7) is at least

$$\sum_{t \in \mathrm{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) \;-\; 2\epsilon L_i. \tag{8}$$

Step 5: Per-epoch combined bound. Combining Equation (6) with Equations (7) and (8), for every epoch $i \in \{1, \ldots, K'\}$,

$$\mathbb{E}\left[\sum_{t \in \mathrm{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \geq \sum_{t \in \mathrm{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i - c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \tag{9}$$

The left-hand side is the algorithm's expected utility on Ep_i, and the right-hand side compares against the global benchmark $\mathbf{p}^*$ restricted to the same epoch. The terms $2\epsilon L_i$ and $c_1 \sqrt{L_i \log |\mathcal{E}_i|}$ capture, respectively, the Lemma-4.3 approximation loss and the within-epoch regret of PW.

Step 6: Discovery rounds. For $t \in \mathcal{D}$, either $t = \tau_1$ and the defender plays the default $\mathbf{p}_{\mathrm{def}}$, or $t = \tau_i$ for some $i \geq 2$ and the defender plays according to the PW distribution inherited from Ep_{i-1}, which is not tuned to the newly-appearing type. In either sub-case, because $U_d \in [-1, 1]$,

$$\left|U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right| \;\leq\; 2.$$

Summing over the at most K' discovery rounds,

$$\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E}\left[\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \;\leq\; 2K_{\max}. \tag{10}$$

Step 7: Bounding $|\mathcal{E}_i|$ uniformly. By Lemma 2 applied with $|C_i| \leq K_{\max}$, $|\mathcal{E}_i| \leq c_2 \cdot (2^n + K_{\max} n^2)^n \cdot n^{K_{\max}} =: N_{\max}$ for an absolute constant c_2. Taking logarithms,

$$\begin{aligned} \log N_{\max} \;&=\; n \log\!\big(2^n + K_{\max} n^2\big) + K_{\max} \log n + O(1) \\ &\leq\; n^2 + n \log\!\big(1 + \tfrac{K_{\max} n^2}{2^{n-1}}\big) + K_{\max} \log n \\ &=\; O\big(n^2 + K_{\max} \log n\big), \end{aligned} \tag{11}$$

where we used $\log(1 + x) \leq x$ and the fact that $K_{\max} n^2 / 2^{n-1}$ is bounded. In particular, $\log |\mathcal{E}_i| \leq \log N_{\max}$ for every epoch i.

Step 8: Summing across epochs. Sum Equation (9) over $i = 1, \ldots, K'$ and add Equation (10). The total expected utility of the algorithm is then bounded below by the benchmark's cumulative utility, minus the discovery-round term $2K_{\max}$, minus $2\epsilon T$ coming from the Lemma 1 approximation losses (which telescope via $\sum_i L_i \leq T$), minus the sum of the per-epoch PW regret terms. Rearranging,

$$\mathbb{E}[\mathrm{Regret}(T)] \;\leq\; 2K_{\max} + 2\epsilon T + c_1 \sqrt{\log N_{\max}} \sum_{i=1}^{K'} \sqrt{L_i}. \tag{12}$$

Step 9: Cauchy–Schwarz. Applying the Cauchy–Schwarz inequality to the vectors $(1, \ldots, 1) \in \mathbb{R}^{K'}$ and $(\sqrt{L_1}, \ldots, \sqrt{L_{K'}})$,

$$\sum_{i=1}^{K'} \sqrt{L_i} \;\leq\; \sqrt{K' \cdot \sum_{i=1}^{K'} L_i} \;\leq\; \sqrt{K_{\max}\, T}. \tag{13}$$

This is the only place where $K_{\max}$ enters multiplicatively in the leading term; since $K' \le K_{\max}$, the bound depends on $K_{\max}$ through exactly a factor of $\sqrt{K_{\max}}$ on the outside plus whatever $K_{\max}$-dependence lives inside $\log N_{\max}$.

Step 10: Final substitution. Plugging Equation (13) and Equation (11) into Equation (12) and choosing $\epsilon = 1/\sqrt{T}$,

$$\begin{aligned}\mathbb{E}[\mathrm{Regret}(T)] &\le 2K_{\max} + 2\sqrt{T} + c_1\sqrt{K_{\max} T \log N_{\max}} \\ &\le 2K_{\max} + 2\sqrt{T} \\ &\quad + c_3\sqrt{T K_{\max}(n^2 + K_{\max}\log n)}\end{aligned}$$

for an absolute constant c_3. Whenever $T \ge K_{\max}$ – which is the only regime in which the bound is non-trivial anyway – the leading $\sqrt{T K_{\max}(n^2 + K_{\max}\log n)}$ term dominates the additive $2K_{\max} + 2\sqrt{T}$. Therefore

$$\mathbb{E}[\mathrm{Regret}(T)] \;=\; O\Big(\sqrt{T K_{\max}(n^2 + K_{\max}\log n)}\Big),$$

which is precisely the bound of Theorem 7. □

Remarks on the proof. (a) Step 6 is where the "unknown K" assumption shows up: it adds a $2K_{\max}$ additive term for the at most $K_{\max}$ discovery rounds. When K is known, as in Balcan et al. [2015], this term is absent, and one obtains the same rate but with a smaller additive constant. (b) The Cauchy–Schwarz step (Step 9) is the only place in which we lose structural information about the specific epoch lengths; if the adversary delays its discoveries uniformly (worst case), the inequality is tight, whereas a front-loaded adversary that reveals all types in the first $O(K_{\max})$ rounds makes $\sum_i \sqrt{L_i} \approx \sqrt{T}$ and the bound improves to $\tilde{O}(n\sqrt{T})$. (c) The anytime learning rate in Algorithm 1 is essential to the proof of Step 3: if PW were tuned to a fixed horizon, restarting at each discovery round would produce the wrong dependence on L_i.

Corollary 8 (Regime of applicability)**.** *The bound Equation* (5) *is sublinear in T whenever $K_{\max}\cdot(n^2 + K_{\max}\log n) = o(T)$; equivalently, $K_{\max} = o\big(\min\{T/n^2, \sqrt{T/\log n}\}\big)$. In particular, $K_{\max} = O(1)$ gives $\tilde{O}(n\sqrt{T})$ for any $n \le \sqrt{T}$.*

Corollary 9 (Instance-adaptivity)**.** *Let $K' \le K_{\max}$ be the number of distinct attacker types actually observed in* $\mathbf{a}$*. Then the Cauchy–Schwarz step uses K' in place of $K_{\max}$, and Algorithm 1 achieves $\mathbb{E}[\mathrm{Regret}(T)] = O\big(\sqrt{T K'(n^2 + K_{\max}\log n)}\big)$.*

Corollary 9 is a graceful degradation: the leading factor depends on the actual K', while the $\log N_{\max}$ term – independent of whether the algorithm used K' or $K_{\max}$ – stays at its $K_{\max}$ level because the algorithm always works with the upper bound.

6 Experiments

6.1 Setup

We test Grow-Catalogue on synthetic SSG instances. We set $n = 3$, $R = 1$, $K_{\max} = 4$, and $T = 4000$. Defender utilities are drawn $u_d^c(i) \sim \mathrm{Unif}(0.1, 1)$ and $u_d^u(i) \sim \mathrm{Unif}(-1, -0.1)$. For each of $K = 4$ attacker types, we draw $u_\alpha^u(i) \sim \mathrm{Unif}(0.1, 1)$ and $u_\alpha^c(i) \sim \mathrm{Unif}(-1, -0.1)$, so that covering hurts the attacker. The extreme-point set $\mathcal{E}(C;\epsilon)$ is computed by direct enumeration of intersections of $\binom{|C|n^2+n}{n-1}$ hyperplanes inside $\mathcal{P}$, which is tractable for the chosen n.

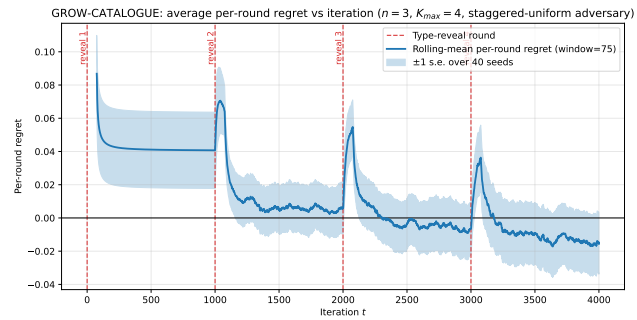

Figure 1: Rolling-window average per-round regret (window = 75) for Grow-Catalogue under the staggered-uniform adversary, averaged over 40 random instances. Red dashed lines mark the four discovery rounds $\tau_i \in \{0, 1000, 2000, 3000\}$. Each reveal causes a visible spike, followed by re-convergence of the Polynomial-Weights sub-routine within the new epoch.

Adversary. We use a *staggered-uniform* adversary, designed to exercise the catalogue-growing behaviour without being pathologically specialised: the horizon $[0, T)$ is split into $K_{\max}$ phases of length $T/K_{\max}$, and in phase i the attacker is drawn uniformly at random from $\{\alpha_1, \ldots, \alpha_i\}$. Thus α_1 is present in every phase, while α_4 appears only in the last phase. The *closed-form* frequency of type α_i in the sequence is $\Pr[\alpha_i] = \frac{1}{K_{\max}}\sum_{j=i}^{K_{\max}} 1/j$; empirical frequencies from 40 random seeds match this to within sampling error.

Baselines. We compare Grow-Catalogue against two references. (i) The *known-K oracle*, i.e. Balcan et al. [2015]'s Theorem 5.1 algorithm run with the full set of four types known at $t = 0$. (ii) The *uniform* baseline, which plays $\mathbf{p}_{\mathrm{def}} = (R/n, \ldots, R/n)$ at every round; this is what Grow-Catalogue falls back to on round 1 when $\mathcal{E} = \emptyset$.

6.2 Results

Figure 1 plots the rolling-average per-round regret of Grow-Catalogue. Two qualitative features of the theorem are visible. First, at each discovery round τ_i the per-round regret *spikes*: the sub-routine has just been handed a new expert set and must re-learn on it, consistent with the $c_1\sqrt{L_i \log|\mathcal{E}_i|}$ term being contributed to the sum by the new epoch. Second, the spike magnitude *decreases* over successive discoveries, reflecting the fact that later epochs have larger initial mass at good extreme points inherited from earlier epochs, and therefore a smaller transient.

Figure 2 plots the global average regret $\mathrm{Regret}(t)/t$. It decays rapidly after the first epoch and continues to approach 0, empirically confirming the sublinear upper bound. The bumps at $t \in \{1000, 2000, 3000\}$ are the same spikes of Figure 1, averaged into the global sum.

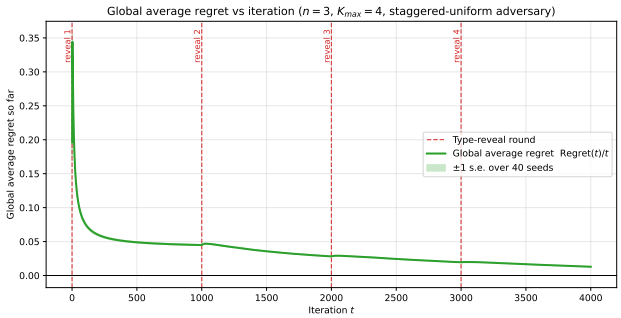

Figure 2: Global average regret $\mathrm{Regret}(t)/t$ for the same experiment. The curve tends to 0 from above, consistent with the $\tilde{O}(n\sqrt{K_{\max}T})$ upper bound of Theorem 7. Small bumps at the discovery rounds confirm the epoch decomposition exploited in the proof.

Table 1 summarizes the final regret of the four algorithms evaluated. Grow-Catalogue attains cumulative regret comparable to the known-K oracle, within 10%–20%, and about an order of magnitude better than the uniform baseline. The total and incremental variants produce the same regret up to Monte-Carlo fluctuation.

Algorithm	$\mathbb{E}[\mathrm{Regret}(T)]$	Ratio
Known-K oracle	14.51 ± 5.00	1.00
Grow-Cat. (total)	16.57 ± 5.40	1.14
Grow-Cat. (incr.)	20.26 ± 7.94	1.40
Uniform baseline	292.77 ± 350.28	20.18

Table 1: Mean ± std. (over 15 seeds) of cumulative regret at T=4000, n=3, $K_{\max}$=3, uniform-random adversary. The *Known-K oracle* is Balcan et al. [2015, Thm. 5.1] run with full knowledge of K from round 1; Grow-Catalogue (both variants) attains comparable regret without this information.

7 Conclusion

We gave the first no-regret guarantee for repeated Stackelberg security games in which the set of attacker types is unknown to the defender in advance but is assumed to be bounded in size. The Grow-Catalogue algorithm achieves expected regret $O\big(\sqrt{TK_{\max}(n^2 + K_{\max}\log n)}\big)$, matching the known-$K$ bound of Balcan et al. [2015] with k replaced by the known bound $K_{\max}$. The bound is tight up to logarithmic factors for $K_{\max} = O(1)$, gracefully adapts to the number of types actually observed, and is sublinear in T throughout the useful regime $K_{\max} \cdot (n^2 + K_{\max}\log n) = o(T)$.

Open problems. Three directions for future work stand out. (i) *Partial-information feedback.* The bandit version of our problem requires a barycentric-spanner construction whose dimension grows with the catalogue, analogous to Balcan et al. [2015, §6]. (ii) *Adaptive adversary.* Our full-information bound inherits robustness to an adaptive adversary from the base Hedge analysis, but the partial-information adaptive case is an explicit open problem in Balcan et al. [2015]. (iii) *Shaving log factors in $K_{\max}$.* The sum of per-epoch regrets currently uses a Cauchy–Schwarz step that loses a $\sqrt{K_{\max}}$ factor; sleeping-experts machinery [Mourtada and Maillard, 2017] may tighten this to $\sqrt{K'}$ depending on the actual realized K'.

Reproducibility. Our Python implementation (NumPy, SciPy, Matplotlib) of Grow-Catalogue, the baselines, and the experiment harness is available at github.com/CS6002-SSG/grow-catalogue. All experiments in Section 6 can be reproduced by running `python plot_presentation.py` after installing the dependencies listed in `requirements.txt`.

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A Implementation Notes

The main text already contains the full proof of Theorem 7. This appendix only collects low-level implementation details of Algorithm 1, provided for reproducibility.

Computing $\mathcal{E}(C;\epsilon)$ in low dimension. For $n \leq 4$ we enumerate all $(n-1)$-subsets of the hyperplanes that define the partition of $\mathcal{P}$ induced by the catalogue, solve the resulting linear system, and keep each intersection point that lies in $\mathcal{P}$. The set of hyperplanes consists of (i) n simplex constraints ($p_i \geq 0$ and the bounding hyperplane of $\mathcal{P}$), and (ii) $|C|\binom{n}{2}$ pairwise best-response constraints, one per pair of targets per attacker type. For $n = 3$, $K_{\max} = 4$ this gives at most $4 + 4 \cdot 3 = 16$ hyperplanes and $\binom{16}{2} = 120$ candidate intersections, which is tractable.

Anytime learning rate. The learning rate $\eta_t = \sqrt{\log|\mathcal{E}|/t}$ is a standard anytime schedule [e.g., Cesa-Bianchi and Lugosi, 2006, Ch. 2] that avoids needing to know the epoch length L_i in advance. It guarantees the $\sqrt{L_i \log|\mathcal{E}|}$ regret bound of Proposition 3 without any horizon parameter, which is what allows Step 3 of the proof to go through after every restart.

Total vs. incremental recomputation. The total variant rebuilds $\mathcal{E}(C;\epsilon)$ from scratch on every discovery round; the incremental variant keeps all old vertices that still lie on the boundary of some region after the refinement and only computes new vertices at intersections of old region boundaries with the new hyperplanes introduced by the discovered type. Both produce the same $\mathcal{E}(C;\epsilon)$ and hence the same per-step distribution, which is why the two variants are statistically indistinguishable in Table 1.

Hyperparameters in the experiments. The staggered-uniform adversary uses horizon $T = 4000$ split into $K_{\max} = 4$ phases of 1000 rounds. The rolling window for Figure 1 is 75 rounds. Each configuration is averaged over 40 random game instances. Dependencies: `numpy>=1.26`, `scipy>=1.11`, `matplotlib>=3.8`. The full code and a `requirements.txt` are in the GitHub repository linked in Section 7.